

Corrective Action Report — CAR-2025-009

Field	Value
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CAR Status	Open
Subject	Repeat-Failure Assets Not Escalated to Engineering Review per Policy ER-3; WIN-10131, WIN-10016, SUB-10641 Demonstrate Sustained Non-Compliance
Affected Region / Site / Asset	NW-PowerPool / NW-PowerPool-WIND-014, NW-PowerPool-WIND-024 / WIN-10131, WIN-10016, SUB-10641 (primary); WIN-10011, WIN-10058 (secondary)
Severity	Critical
Owner	E. Linhart, Principal Engineer — Rotating Equipment; co-owner T. Harlan, Director of Reliability NW-PowerPool
Original Target Close Date	2026-03-31
Distribution	VP Operations; Engineering Standards Committee; NW-PowerPool Plant Manager; Asset Management; Internal Audit

1. Finding

Engineering Review Policy **ER-3** ("Repeat-Failure Asset Escalation"), revised 2023-09-12, requires that any asset accumulating three (3) forced outages within a rolling 12-month window be referred to formal Engineering Review within 30 days of the third event. Engineering Review produces an Asset Disposition Recommendation (continued operation, refurbish, replace, retire) which becomes the controlling planning document for that asset.

Audit of the Engineering Review queue and CMMS outage ledger confirms that three of the most repeat-failed assets in the Cascadia fleet were either escalated late or have never been formally reviewed:

- **WIN-10131** (Siemens Gamesa, NW-PowerPool-WIND-024) — third forced outage occurred on 2025-04-09. ER-3 escalation due 2025-05-09. **Not escalated until 2025-09-22** (after the 8th outage). Asset Disposition Recommendation issued 2025-11-10 as "Pending — gearbox replacement decision contested" (cross-reference CAR-2025-002).
- **WIN-10016** (GE, NW-PowerPool-WIND-014) — third forced outage occurred on 2024-11-18 (the asset began the year already in violation of ER-3). 16 outages cumulative; ER-3 escalation **never formally filed**. Engineering has been engaged on the asset informally but no Asset Disposition Recommendation exists.
- **SUB-10641** (ABB, NW-PowerPool-WIND-014) — third forced outage occurred on 2025-03-04. ER-3 escalation due 2025-04-03. **Added to "advisory list" 2025-09-09 only** (cross-reference CAR-2025-005 §6), which is not the same as formal Engineering Review. 15 outages cumulative; no Asset Disposition Recommendation.

2. Evidence

2.1 ER-3 escalation timing audit

Asset	3rd Forced Outage Date	ER-3 Escalation Due	Actual Escalation	Latency	Outages at Escalation
WIN-10131	2025-04-09	2025-05-09	2025-09-22	+136 days	8
WIN-10016	2024-11-18	2024-12-18	Not filed	>365 days	16 (cumulative)
SUB-10641	2025-03-04	2025-04-03	Advisory only (2025-09-09); not formal	>180 days	15 (cumulative)
WIN-10011	2025-03-20	2025-04-19	2025-10-14	+178 days	9 (at escalation)
WIN-10058	2025-05-02	2025-06-01	2025-08-26	+86 days	7 (at escalation)

2.2 Counter-examples (compliant)

- **WIN-10130** (Siemens Gamesa, CAISO-WIND-026) — escalated 2025-03-12 after 3rd outage 2025-02-11. Latency +29 days, within policy.
- **TRA-10570** (Siemens, NYISO-SOLAR-031) — escalated within 14 days of 3rd outage.

Compliance gap is concentrated in NW-PowerPool. CAISO, NYISO, and PJM have demonstrated ER-3 compliance latency under 30 days for all FY2025 escalations.

2.3 Financial impact of late escalation

The five NW-PowerPool assets above accumulated **64 forced outages** between them in FY2025 against **15** that would have been the expected outage count had Engineering Review intervened at the 3rd event and prescribed disposition within 60 days. The differential (49 forced outages) maps to approximately **18,400 MWh of lost generation** valued at approximately **\$1.1M** in revenue, plus emergency-vendor invoices of approximately **\$3.8M** (cross-reference CAR-2025-004 invoice schedule).

If ER-3 had been complied with, the Period B Fall Overheating Sequence is unlikely to have unfolded as it did — WIN-10131 would have been under disposition by August 2025 rather than in dispute, and SUB-10641 would have been under formal review by spring.

2.4 The "advisory list" workaround

Engineering at NW-PowerPool has been operating an internal "advisory list" of repeat-failure assets that receives ad-hoc engineering attention but does not produce a formal Asset Disposition Recommendation. The advisory list contains 11 assets including the five named in §2.1. The advisory list is not recognized under ER-3; its existence has the effect of making ER-3 violations invisible to the formal Engineering Review queue.

3. Root Cause Hypothesis

- **Primary:** No CMMS-integrated trigger exists to enforce the ER-3 30-day window. The policy requires manual filing by site reliability personnel.
- **Contributing:** Site reliability personnel at NW-PowerPool reportedly understood the advisory list as satisfying ER-3, an understanding that was never formally challenged.
- **Contributing:** The Engineering Review queue has been understaffed in 2025 (one engineer departure not backfilled). Even formally filed escalations have run at ~75-day average time-to-recommendation, which itself exceeds the policy expectation of 60 days.

- **Contributing:** The interaction with CAR-2025-002 (MOC-2025-022 pending) created a downstream paralysis: even when WIN-10131 was eventually escalated, the disposition recommendation could not be finalized without the threshold dispute being resolved.

4. Conflict With Existing Guidance

- **ER-3 §3.1** requires formal escalation; the advisory list is not a recognized substitute.
- **CAR-2024-118** ("Preventive Maintenance Cadence Inadequate For Aging Wind Assets," issued December 2024) named WIN-10016 and SUB-10641 explicitly; remediation of CAR-2024-118 should have included ER-3 escalation. CAR-2024-118 remains open.
- **MS-2025-03 §2.4** requires that Engineering Review Recommendations control the operating envelope of an asset. Operating an asset outside formal review is, strictly read, an MS-2025-03 violation.

5. Recommended Corrective Action

1. Immediate (30-day) actions:

2. File formal ER-3 escalation for WIN-10016, SUB-10641, WIN-10011, WIN-10058 within 30 days of CAR re-issue. WIN-10131 escalation already filed.
3. Retire the "advisory list" as a category. Migrate its contents into the formal Engineering Review queue.

4. Structural actions:

5. Modify CMMS to auto-file ER-3 escalation on the day a third forced outage closes. Manual cancellation requires Engineering Standards Committee approval.
6. Backfill the open Engineering Reviewer position. Reassess queue capacity against caseload.
7. Establish 60-day SLA on Asset Disposition Recommendation from filing date.
8. Add ER-3 compliance to the Director of Reliability NW-PowerPool annual performance review.

9. Asset-specific actions:

10. WIN-10016: Engineering Review to address VTM-GE-2023 manual obsolescence (CAR-2025-007) as part of disposition.
11. SUB-10641: Engineering Review to address MS-2025-03 §8.1 cycle conflict (CAR-2025-005).

12. WIN-10131: Engineering Review pending resolution of MOC-2025-022 (CAR-2025-002).

6. Action Owner & Timeline

Sub-action	Owner	Target Date
File ER-3 escalations for 4 named assets	T. Harlan	2026-01-15
Retire advisory list / migrate to queue	E. Linhart	2026-01-31
CMMS auto-file modification	R. Vasquez	2026-03-31
Engineering Reviewer position filled	E. Linhart + HR	2026-02-28
60-day SLA standing order	Engineering Standards Committee	2026-02-28
Asset Disposition Recommendations (5 assets)	E. Linhart	2026-04-30

7. Implementation Status

Open — Critical. No CMMS modification has been deployed. The advisory list remains active. The Engineering Reviewer position has been posted but not filled. WIN-10131 Asset Disposition Recommendation has been issued in "Pending" status only and is gated on resolution of CAR-2025-002. Forced outages on WIN-10016 and SUB-10641 have continued post-CAR (OUT-2025-1632 occurred 2025-11-18). The cumulative pattern is now visible enough that it is being included in the FY2025 Reliability Annual Report under "Material Control Weakness."

Cross-references: ER-3 (Engineering Review Policy, internal); CAR-2024-118 (predecessor PM-cadence finding naming WIN-10016 and SUB-10641); CAR-2025-001 (deferred PMs upstream of repeat-failure pattern); CAR-2025-002 (MOC-2025-022 paralysis affecting WIN-10131 disposition); CAR-2025-005 (SUB-10641 maintenance-cycle conflict); CAR-2025-007 (VTM-GE-2023 manual obsolescence affecting WIN-10016 disposition); MS-2025-03 §2.4; OIR-2025-003 (Period A); OIR-2025-007 (Period B); EMR-2025-001 through EMR-2025-006 (emergency invoices for these same assets); RCA-2025-001 through RCA-2025-005.

End of report.